DAY 31 | HANDS-ON ACTIVITIES

GenAI System Evaluation: Problem Statements

Eight problems on evaluating the policy assistant and the SmartDesk agent. Offline metrics are simplified stand-ins: they show how the code works, not real model quality. With a key, use LLM judges and real embeddings.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Write the Eval Set

Level: Warm-up | **Concepts:** dataset design

Write 12 questions with an expected key phrase, the source sentence, a topic tag and a language tag. Include 3 Japanese questions and 2 unanswerable ones.

Sample run / expected result

```
{"q": "How many sick days?", "expect": "8 days", "source": "8 days of paid sick lea
```

Hints

- Use the words real users would use, not the document's words.

Problem 2: Retrieval Metrics

Level: Easy | **Concepts:** hit rate, MRR

Compute hit@1, hit@3 and MRR for your eval set.

Sample run / expected result

```
hit@1 0.67 | hit@3 0.83 | MRR 0.74
```

Hints

- MRR = mean of 1 / rank of the first correct result.

Problem 3: Faithfulness

Level: Medium | **Concepts:** claim checking

Compute faithfulness for each answer. Then use the planted hallucinating generator and show the score drop.

Sample run / expected result

```
Faithful generator: 0.96
Hallucinating generator: 0.52
```

Hints

- Split answers into sentences; check each against the retrieved context.

Problem 4: Error Analysis and Slices

Level: Medium | **Concepts:** retrieval miss vs generation error, bias

Classify every failure as retrieval miss or generation error, then report correctness by language and by topic. Recommend the first fix.

Sample run / expected result

```
retrieval_miss: 3 (mostly Japanese)
generation_error: 1
en: 0.89 | ja: 0.00 (offline)
First fix: multilingual embeddings
```

Hints

- Read the failures before choosing a fix.

Problem 5: Calibrate a Judge NEW (v4)

Level: Challenging | **Concepts:** rubric, agreement, Cohen's kappa

Write a rubric-based judge (1 to 5 correctness with reasons). Label 10 answers yourself and compute exact agreement, agreement within 1 point and Cohen's kappa on pass/fail.

Sample run / expected result

```
Exact agreement: 6/10
Within 1 point: 9/10
Kappa (pass/fail): 0.58
Judge was lenient on partial answers
```

Hints

- Ask for the reason before the score.
- Kappa corrects for agreement that would happen by chance.

Problem 6: Score an Agent Trajectory NEW (v4)

Level: Medium | **Concepts:** trajectory evaluation

For four SmartDesk agent runs, compare the tool calls with the expected trajectory. Report tool selection, in-order match, extra calls, forbidden calls and final answer correctness.

Sample run / expected result

```
run  tools_ok  in_order  extra  forbidden  answer_ok
R1   True      True      0      0          True
R3   True      False     1      0          True   <- right answer, messy path
R4   False     False     0      1          False  <- called issue_refund
```

Hints

- A correct answer with a forbidden call is still a failure.

Problem 7: pass@k vs pass^k NEW (v4)

Level: Medium | **Concepts:** reliability, non-determinism

Run each of five tasks k = 1, 3 and 5 times with the notebook's simulated agent. Compute pass@k and pass^k and explain the gap to a customer in two sentences.

Sample run / expected result

```
k=1: pass@k 0.80 | pass^k 0.80
k=5: pass@k 1.00 | pass^k 0.40
```

Hints

- pass@k is what demos show; pass^k is what customers feel.

Problem 8: Security Regression Eval NEW (v4)

Level: Stretch | **Concepts:** safety metrics per version

Run the Day 30 attack and benign sets against two guardrail versions. Report block rate, false alarm rate and forbidden-call rate for each and flag any regression.

Sample run / expected result

```
guard v1: block 0.90 | false alarms 0.10
guard v2: block 0.70 | false alarms 0.00  <- REGRESSION
```

Hints

- A version that blocks less but annoys fewer users can still be unsafe.